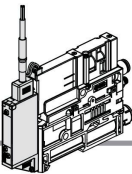


ZK2 Series

Digital Pressure Switch for Vacuum Specifications

Digital pressure switch
for vacuum with energy
saving function



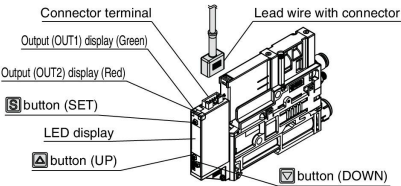
Digital Pressure Switch for Vacuum Ejector with Energy Saving Function

Model		Specifications
Rated pressure range		-100 to 100 kPa
Set pressure range		-105 to 105 kPa
Proof pressure		500 kPa
Smallest settable increment		0.1 kPa
Applicable fluid		Air/Non-corrosive gas/Incombustible gas
Power supply voltage		12 to 24 VDC $\pm 10\%$ Ripple (P-P) 10% or less (Protected against reverse connection)
Current consumption		40 mA or less
Switch output		NPN or PNP open collector OUT1: General purpose, OUT2: Valve control
	Maximum load current	80 mA
	Maximum applied voltage	26.4 VDC
	Residual voltage	2 V or less (with load current at 80 mA)
	Response time	2.5 ms or less (Anti-chattering function working: 20, 100, 500, 1000 or 2000 ms selected)
	Short circuit protection	Yes
Repeatability		$\pm 0.2\%$ F.S. ± 1 digit
Hysteresis	Hysteresis mode	Variable (0 or above) Note)
Display		3 1/2 digit, 7-segment LED, 1-color display (Red)
Display accuracy		$\pm 2\%$ F.S. ± 1 digit (Ambient temperature at 25 $\pm 3^\circ\text{C}$)
Indicator light		Lights up when output is turned ON. OUT1: Green, OUT2: Red
	Enclosure	IP40
Environmental resistance	Operating humidity range	5 to 50°C
	Withstand voltage	1000 VAC for 1 minute between terminals and housing
	Insulation resistance	50 M Ω or more (500 VDC measured via megohmmeter) between terminals and housing
Temperature characteristics		$\pm 2\%$ F.S. (at 25°C in an operating temperature range of 5 and 50°C)
Lead wire		Cable: 5 cores $\phi 3.5$, 2 m Cross section: 0.15 mm ² (AWG26) Insulator O.D.: 1.0 mm
Standards		CE marking, RoHS

Note) If the applied pressure fluctuates around the set value, the hysteresis must be set to a value more than the fluctuating width, otherwise, chattering will occur.

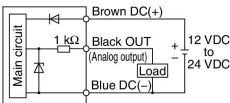
Description (Pressure Switch for Vacuum)

Output (OUT1) display (Green)	Lights up when OUT1 is turned ON.
Output (OUT2) display (Red)	Lights up when OUT2 is turned ON.
LED display	Displays the current pressure, set mode and error code.
▲ button (UP)	Selects the mode or increases the ON/OFF set-value. Use for switching to the peak display mode.
▼ button (DOWN)	Selects the mode or decreases the ON/OFF set-value. Use for switching to the bottom display mode.
Ⓢ button (SET)	Use for changing the mode or setting the set-value.



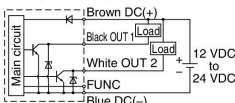
Internal Circuit and Wiring Example

■ Pressure Sensor ZK2-PS□-A



Voltage output type: 1 to 5 V
Output impedance: Approx. 1 k Ω

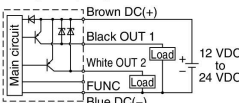
■ Pressure Switch for Vacuum ZK2-ZS□A□□-A NPN (2 Outputs)



Max. 28 V, 80 mA
Residual voltage: 2 V or less

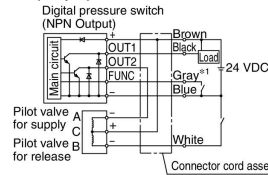
*The FUNC terminal is connected when using the copy function. (Refer to the Operation Manual.)

ZK2-Z□B□□-A PNP (2 Outputs)

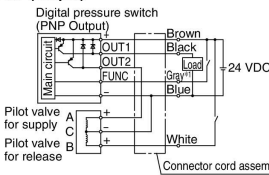


Max. 80 mA
Residual voltage: 2 V or less

■ Pressure Switch for Vacuum with Energy Saving Function ZK2-ZSVA□□-A NPN (Output)



ZK2-ZSVB□□-A PNP (Output)



*1 The gray wire (FUNC) is connected when operating the supply valve by energy-saving control (for workpiece adsorption). (Refer to the Operation Manual.)